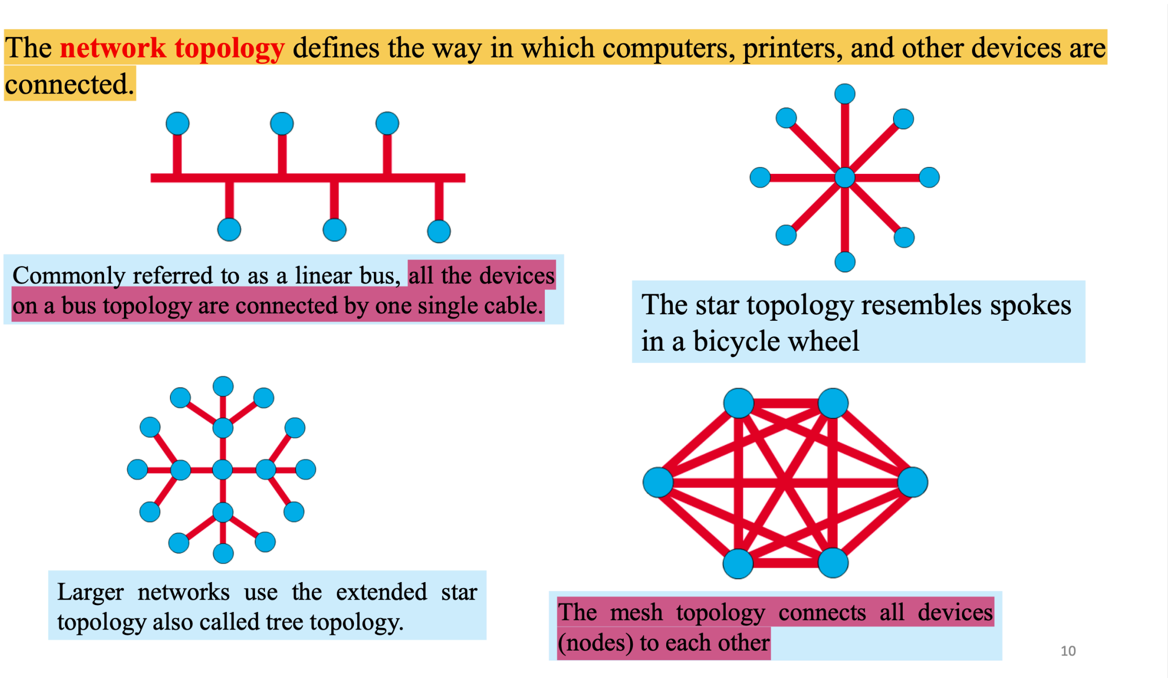
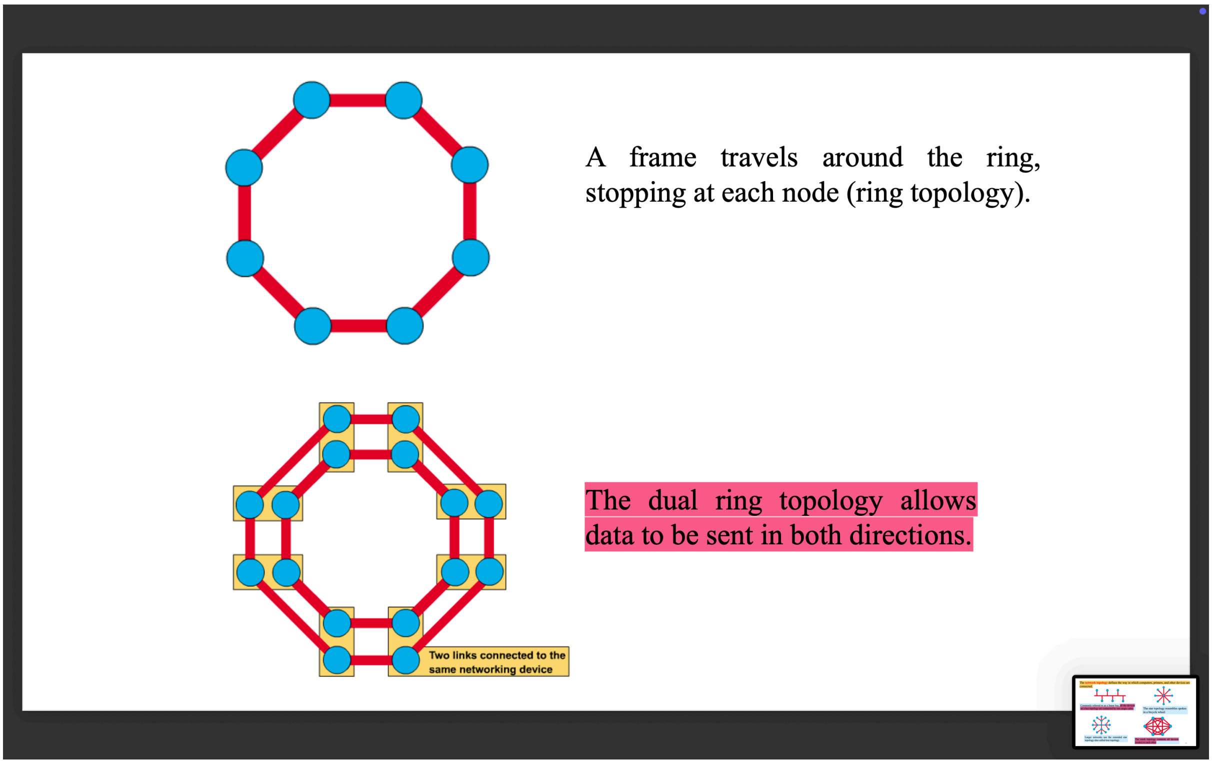
List of Question of slide - 1

1. Draw different topology of computer network (p: 10-11)





for better understand:

Here's a clear explanation of different **network topologies**, which define how devices (nodes) are connected in a computer network:

🌐1. Bus Topology

- **Structure**: All devices share a **single communication line** (the bus).
- **Working**: Data travels in both directions and devices listen for data addressed to them.

✅ **Advantages**:

- Easy to set up
- Requires less cable

❌ **Disadvantages**:

- Entire network fails if the bus fails
- Performance drops with more devices

🕸️2. Star Topology

- **Structure**: All devices connect to a **central hub or switch**.
- **Working**: Data is sent from the source to the hub, which then forwards it to the destination.

✅ **Advantages**:

- Easy to add/remove devices
- If one device fails, the network continues to work

❌ **Disadvantages**:

- Hub failure brings down the entire network
- More cabling required

🔁3. Ring Topology

- **Structure**: Devices are connected in a **circular loop**, each having exactly two neighbors.

- **Working**: Data travels in one direction (or two in dual ring); each node passes data along the ring.

✅ **Advantages**:

- Predictable performance with few collisions
- Easier to manage with token-based systems

❌ **Disadvantages**:

- Break in the ring affects the whole network
- Troubleshooting can be complex

🔗4. Mesh Topology

- **Structure**: Each device is **connected to every other device** (full or partial).
- **Working**: Offers multiple paths for data to travel.

✅ **Advantages**:

- High fault tolerance
- Reliable and secure

❌ **Disadvantages**:

- Very expensive and complex to install
- Requires a lot of cabling

🌲5. Tree Topology (Hierarchical)

- **Structure**: A hybrid of star and bus topologies. It has a root node and branches in a hierarchical manner.
- **Working**: Central nodes manage traffic and distribute it to lower levels.

✅ **Advantages**:

- Scalable and structured
- Easy fault detection

❌ **Disadvantages**:

- Central node failure impacts the whole tree
- Requires more maintenance

🔄6. Hybrid Topology

- **Structure**: Combines two or more different topologies (e.g., star-ring, star-bus).
- **Working**: Inherits features of its constituent topologies.

✅ **Advantages**:

- Flexible and scalable
- Can be tailored to meet specific needs

❌ **Disadvantages**:

- Complex and expensive
- Hard to manage

Summary Table:

Topology	Central Device	Fault Tolerance	Cabling Cost	Scalability
Bus	No	Low	Low	Low
Star	Yes	Medium	Medium	High
Ring	No	Low	Medium	Medium
Mesh	No	High	High	Medium
Tree	Yes	Medium	High	High
Hybrid	Varies	High	High	High

Would you like a diagram showing these topologies visually?

2. Draw a three layer protocol stack of postal mail service (Page- 15)

Example-1

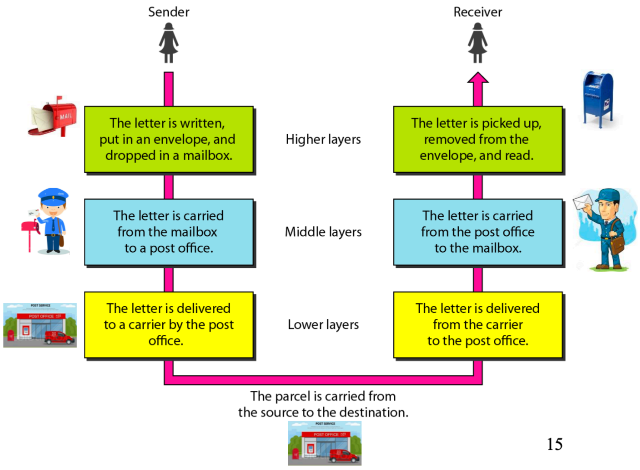
We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.

Sender Side

Higher layer: The sender writes the letter, inserts in an envelope, writes the sender and receiver addresses, and drops in a mailbox.

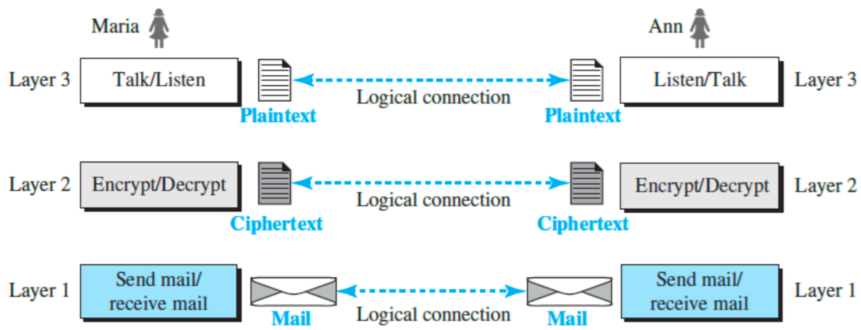
Middle layer: The letter is picked up by a letter carrier and delivered to the post office.

Lower layer: The letter is sorted at the post office and deliver to the carrier transports.



15

3. Draw a 3 layer protocol stack of showing logical connection / peer to peer communication (Page - 21)



Layer-to-layer virtual communication is considered as logical connections between each layer as shown in Figure above.

21

For better understanding

🧠 What is Logical Connection in Computer Networks?

A **logical connection** refers to a **virtual path or relationship** between two devices that communicate over a network — even if they are not directly physically connected.**in logical communication, each corresponding layer performs related but not exactly the same tasks.**

💡 Key Points:

- **Virtual Relationship**: It enables two devices (sender and receiver) to communicate **as if they are directly connected**, even though the actual path may pass through several intermediate devices.
- **Established by Protocols**: Logical connections are established using **network protocols** (like TCP) which manage addressing, data integrity, and order.
- **Independent of Physical Path**: The **physical medium** (e.g., fiber, copper, wireless) may change, but the logical connection **remains stable** during the session.

🧩 Example:

In **TCP/IP networks**:

- TCP (Transmission Control Protocol) creates a **logical connection** between two systems by:
  1. **3-way handshake** (SYN, SYN-ACK, ACK)
  2. Ensuring reliable data transfer
  3. Maintaining session until communication ends

🆚 Logical vs Physical Connection

Feature	Logical Connection	Physical Connection
Definition	Virtual communication path	Actual wires, cables, signals
Managed by	Software/Protocols (e.g., TCP)	Hardware (NICs, cables, switches)
Flexibility	Can reroute dynamically	Fixed physical path
Dependency	Independent of actual cabling	Depends on physical layout

---

Would you like an illustration showing logical vs. physical connections?

4. Mention 2 basic Function of Physical Layer (p: 29 → 2,3 point)

Here are **2 basic functions of the Physical Layer** in a computer network:

1. **Bit Transmission**:

   It is responsible for the transmission and reception of raw binary data (0s and 1s) over a physical medium such as a cable or wireless signal.

2. **Physical Medium Interface**:

   It defines the hardware means of sending and receiving data on a carrier, including cables, connectors, voltage levels, and pin layout.

5. Compare Simplex, half, and full duplex

🔁Comparison: Simplex, Half Duplex, and Full Duplex Communication

Mode	Definition	Data Flow Direction	Example
Simplex	Data flows in **only one direction**	One-way only	Keyboard → Monitor
Half Duplex	Data flows **both directions**, but **one at a time**	Two-way (alternate)	Walkie-Talkie, Police Radio
Full Duplex	Data flows in **both directions at the same time**	Two-way (simultaneous)	Phone Call, Video Conferencing

---

✅Summary:

- **Simplex**: One-way communication (Sender → Receiver)
- **Half Duplex**: Two-way, but only one can talk at a time
- **Full Duplex**: Two-way, both can talk and listen simultaneously

Let me know if you'd like a diagram to visualize this!

6. Compare Error detection e error correction (P-32)

Error Detection:

- Error detection involves identifying if any error has occurred during data transmission.
- A special error-checking code (like a checksum or CRC) is added to the frame by the sender.
- The receiver checks this code to detect errors.
- If an error is found or the frame is lost, the receiver requests a retransmission by sending a negative acknowledgement (or no acknowledgement), and the sender resends the frame.

Error Correction:

- Error correction not only detects errors but also fixes them without needing retransmission.
- This is useful in multimedia applications (like images, videos, voice) where real-time delivery is important.
- The receiver uses additional redundant bits (added by the sender) to identify and correct errors on its own.

7. Define flow control (P-33 (1,2) para).

✅Flow Control — Definition (Simplified and Organized)

Flow Control is a technique used in data communication to ensure that the **sender does not overwhelm the receiver** with too much data too quickly.

🔷Key Points:

- Devices in a network often work at **different speeds**.
- Flow control manages **data transmission rate** so that a **fast sender** doesn't flood a **slow receiver**.
- It uses **buffers** and **acknowledgement mechanisms** to regulate the flow.

🔄In Case of Packet Loss or Delay:

- If the sender doesn't receive an **acknowledgement (ACK)** due to noise or errors, it **retransmits** the same packet.
- This can cause **duplicate packets** at the receiver side.

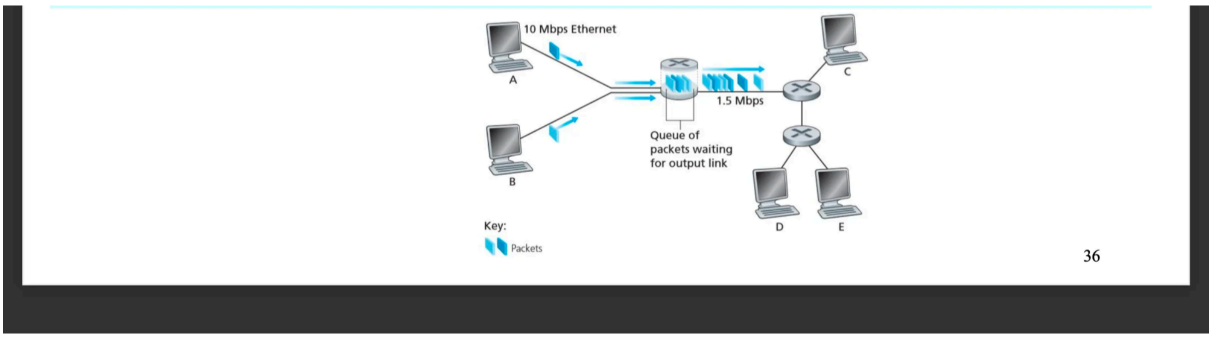
- **Flow control mechanisms handle this issue** to avoid confusion or data corruption.

8. Mention 3 basic function of network layer (36P)

At this layer, the unit of data exchanged among nodes is typically called a packet rather
than a frame, although they are fundamentally the same thing.

✅Three Basic Functions of the Network Layer

1. **Routing of Packets**
   - The network layer determines the **optimal path** for data packets to travel from the source to the destination using **routing tables** maintained by routers.
2. **Traffic Congestion Control**
   - It manages **network congestion** by controlling the flow of data, ensuring smooth delivery, and preventing packet loss due to overloaded network paths.
3. **Quality of Service (QoS) Management**
   - It ensures desired levels of **performance**, including delay, jitter (variation in delay), and bandwidth. The **sum of delays** experienced by a packet is referred to as **latency**.



9. Give the concepts of checkpoints of session Layer (P-45, first 2 para with diagram)

🔷 What is Checkpointing?

- **Session layer provides Checkpoints,** are special **synchronization markers** placed in a continuous data stream.
- They allow the communication session to **resume from the last checkpoint** if the connection is lost due to a failure or crash.

🔁Why Are Checkpoints Important?

1. 🔌 **Recovery from Failure**:

   If a host disconnects in the middle of a long file transfer or data exchange, it can **resume from the last checkpoint** rather than starting over.

2. ⏳ **Efficient Long Transmissions**:

   Checkpointing helps in maintaining the **efficiency** of long-running sessions, especially in unstable networks.

3. 💡 **Reduced Redundancy**:

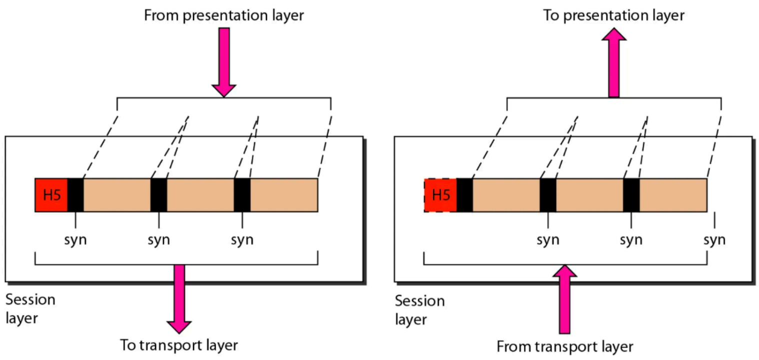
   Avoids retransmitting the **entire data**, saving time and resources.

🧠 Note:

> Checkpointing is **not commonly used OSI-based layered communication systems reliable data applications**

Would you like a simple diagram to visualize how checkpointing works?

The session layer provides: checkpoints or synchronization points to a stream of data, which is not usually used in the Internet Protocol Suite. During long transmissions sometimes a host becomes disconnected in the midst of communication then checkpointing allow them to continue from where they were crashed.



10. Give different. formats of data used in presentation Layer (P_46, 1,2 para)

✅Different Formats of Data Used in the Presentation Layer

The **Presentation Layer (Layer 6)** of the OSI model is responsible for ensuring that data is in a **readable and usable format** for both sender and receiver applications. It acts as a translator between data formats used by the network and the applications.

🔷 Key Functions of the Presentation Layer:

1. **Data Formatting / Translation**

   Converts data into a standard, agreed-upon format for transmission (and back again for the receiver).

2. **Data Compression**

   Reduces the number of bytes needed to transmit data (e.g., JPEG, MPEG, MP3).

3. **Data Encryption & Decryption**

   Ensures secure transmission by encrypting outgoing data and decrypting incoming data.

📄Common Data Formats Handled:

Type	Format Example	Purpose
Text Encoding	ASCII-American Standard Code for Information Interchange (7-bit), EBCDIC-Extended Binary Coded Decimal Interchange Code (8-bit)	Interoperable character encoding
Image Format	JPEG, PNG	Efficient image compression
Video Format	MPEG, MP4	Video compression and delivery
Audio Format	MP3, AAC	Music/audio streaming
Data Structures	Integer sizes (e.g., 16-bit, 32-bit, 64-bit), floats	Standard numeric representation

Application Data	JSON, XML	Structured data exchange

🧠 Summary:

The Presentation Layer makes sure that **data is presented in a standardized, understandable, and secure way** for applications at both ends. It enables communication between different systems that might use different data representations.

Would you like a chart showing how the Presentation Layer fits in the OSI model?

11. Give Operating principle of hub. What is Is limitation. (slide2, P-15).

🔌Operating Principle of a Hub

A **hub** is a basic networking device that connects multiple computers or devices in a **LAN (Local Area Network)**. It works at the **physical layer (Layer 1)** of the OSI model.

✅

How It Works (Operating Principle):

1. When a device sends data to the hub, the hub **receives the signal** on one port.
2. The hub then **broadcasts** that data to **all other connected ports**, regardless of the destination.
3. All devices receive the data, but **only the intended device processes it**; others ignore it.

❌

Limitations of a Hub:

1. **No Intelligence**
    - It does **not filter or direct traffic**, leading to unnecessary data flooding.

2. **Collisions**
   - Operates in a **single collision domain**, increasing chances of **data collisions**.
3. **Poor Scalability**
   - Not suitable for large networks due to performance issues.
4. **No Security**
   - Since data is sent to all devices, **any device can intercept any communication**.
5. **Slow Performance**
   - Shares bandwidth among all devices, reducing speed as more devices are added.

✅ In modern networks, **switches** (Layer 2 devices) have largely replaced hubs due to their **efficiency and intelligence**.

Let me know if you want a hub vs switch comparison too!

🔁Comparison of Transmission Line Utilization for Smaller & Larger Frames under Stop-and-Wait Flow Control

🧠

Stop-and-Wait Flow Control

In **Stop-and-Wait**, the sender transmits **one frame** and waits for an acknowledgment **before sending the next**. Only one frame is in transit at a time.

✅How It Works:

1. The sender transmits **one frame at a time**.
2. It **waits for an acknowledgment (ACK)** from the receiver before sending the next frame.
3. If ACK is not received within a timeout period, the sender **resends the frame**.

📉Limitations:

- **Low efficiency**, especially on long-distance/high-speed links (only one frame in flight at a time)
- **Wasted time** during waiting period
- **Slow throughput**

📌When Used:

- **Simple** and easy to implement
- Suitable for **low-speed or short-distance communication**
- Teaching and understanding basic reliable transmission

🪟Sliding Window Protocol

✅How It Works:

1. Allows **multiple frames** to be sent **before receiving ACKs**.
2. Uses a **window size (n)**: the sender can send n frames without waiting.
3. Receiver sends **ACKs** for received frames and **slides the window** forward.
4. Can use **cumulative or selective ACK**.

📉Limitations:

- More **complex** (needs buffers, window management)
- Requires **more memory** and **sequence number handling**

📌

When Used:

- Used in **high-speed or long-delay** networks (e.g., TCP/IP)
- Needed when **efficient bandwidth utilization** is important
- Example: **TCP uses sliding window**

🔁

Comparison Summary

Feature	Stop-and-Wait	Sliding Window
Frame sent at a time	1	Multiple (window size)
Efficiency	Low	High
Complexity	Simple	Complex
Use case	Short links, simple systems	Long-delay, high-speed links (e.g., TCP)
ACK Handling	One ACK per frame	Cumulative or selective ACKs

Would you like a visual diagram showing how each works?